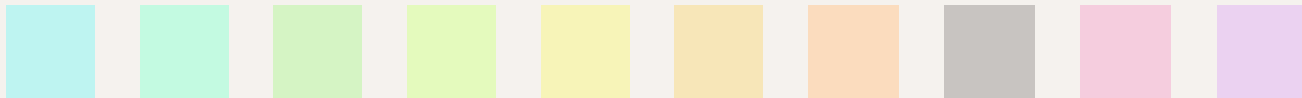


Client Side CSS

מבוא לבניית מערכות ממוחשבות מבוססות אינטרנט הרצאה 3 – CSS



CSS



CSS - Cascading Style Sheets

CSS is a language that describes the style of an HTML document.

What is CSS?

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen
- CSS saves a lot of work by controlling layouts of multiple web pages at once

CSS - Introduction

- HTML was NEVER intended to contain tags for formatting a web page!
- HTML was created to describe the content of a web page
- When tags like `<font>`, and color attributes were added to the HTML, it started a nightmare for web developers.
- Development of large websites, where fonts and color information were added to every single page, became a long and expensive process.
- CSS removed the style formatting from the HTML page!



Inserting CSS

Styling html

- Three Ways to style an html page:
 1. Inline – **style attribute**
 2. Internal – **style tag**
 3. External CSS – **style sheet**

Inline Style Attribute

- Styling an element in the opening tag by using style attribute (what we have done so far...)
- used to apply a unique style to a single HTML element.
- Example:

```
<h1 style="color:blue;">A Blue Heading</h1>  
<p style="color:red;">A red paragraph.</p>
```

A Blue Heading

A Red paragraph

Internal Style Tag

this is a heading

this is a paragraph.

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

External Style Sheet - CSS

- An external style sheet is used to define the style for many HTML pages.
- To use an external style sheet, add a link to it in the <head> section of each HTML page

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

Link CSS to html: <link> Element

- CSS <link> element is added in the <head> tag
- The <link> element dose not have a closing tag
- It contains 2 attributes:
 - Rel- the relationship between the external file and the current file
 - Href- the hyperlink reference to the styles' file

```
<link rel="stylesheet" href="mystyle.css">
```

The Cascade- Cascading Order

- All the styles in a page will "cascade" into the following order:

Inline style (inside an HTML element)



internal style sheets (in the head section)



External style sheets (in the head section)



Browser default



Basic Syntax

CSS Syntax

Selector

Declaration

Declaration

```
h1 {color:blue ; font-size:12px;}
```

↑
Property

↑
Value

↑
Property

↑
Value

CSS syntax example

```
body {  
  background-color: lightblue;  
}
```

```
h1 {  
  color: white;  
  text-align: center;  
}
```

```
p {  
  font-family: verdana;  
  font-size: 20px;  
}
```



Selectors

CSS Selectors

- CSS selectors are used to "find" (or select) the HTML elements you want to style.
- five categories:
 1. Universal (select all elements in the page)
 2. Simple (select elements by name, id, class)
 3. Combinator (select elements by a relationship between them)
 4. Pseudo-class (select elements by a certain state)
 5. Pseudo-elements (select and style a part of an element)
 6. Attribute (select elements by an attribute or attribute value)

Universal Selector

- selects all HTML elements on the page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>selectors</title>
  <style>
    *{
      text-align: center;
      color: blue;
    }
  </style>
</head>
<body>
<div>
  <h1>This is a header</h1>
  <p> this is a paragraph</p>
</div>
<hr>
<div>
  <h2>This is header</h2>
  <p>thi is another paragraph</p>
</div>
</body>
</html>
```

```
* {
  text-align: center;
  color: blue;
}
```

This is a header

this is a paragraph

This is header

thi is another paragraph

Simple Selectors

- By name:

```
p {  
  text-align: center;  
  color: red;  
}
```

- By id:

```
#para1 {  
  text-align: center;  
  color: red;  
}
```

- By class:

```
.center {  
  text-align: center;  
  color: red;  
}
```

Element & Class Selector

- You can also specify that only specific HTML elements should be affected by a class.
- In the example below the selector refers to all p elements with class center

```
p.center {  
  text-align: center;  
  color: red;  
}
```

Grouping Selector

- selects all the HTML elements with the same style definitions

```
h1 {  
  text-align: center;  
  color: red;  
}  
  
h2 {  
  text-align: center;  
  color: red;  
}  
  
p {  
  text-align: center;  
  color: red;  
}
```



```
h1, h2, p {  
  text-align: center;  
  color: red;  
}
```

All Simple Selectors

Selector	Example	Example description
.class	.intro	Selects all elements with class="intro"
#id	#firstname	Selects the element with id="firstname"
*	*	Selects all elements
element	p	Selects all <p> elements
element, element, ..	div, p	Selects all <div> elements and all <p> elements

Combinators Selectors

- A CSS selector can contain more than one simple selector.
Between the simple selectors, we can include a combinator.
- There are four different combinators in CSS:
 - I. descendant selector (space)
 - II. child selector (>)
 - III. adjacent sibling selector (+)
 - IV. general sibling selector (~)

Descendant Selector

- The following example selects all <p> elements inside <div> elements:

```
div p {  
  background-color: yellow;  
}
```

Child Selector

- The child selector selects all elements that are the children of a specified element.
- The following example selects all <p> elements that are children of a <div> element:

```
div > p {  
  background-color: yellow;  
}
```

Adjacent Sibling Selector

- The adjacent sibling selector selects all elements that are the adjacent siblings of a specified element.
- Sibling elements must have the same parent element, and "adjacent" means "immediately following".
- The following example selects all <p> elements that are placed immediately after <div> elements:

```
div + p {  
  background-color: yellow;  
}
```

General Sibling Selector

- The general sibling selector selects all elements that are siblings of a specified element.
- The following example selects all <p> elements that are siblings of <div> elements:

```
div ~ p {  
  background-color: yellow;  
}
```

CSS Pseudo-classes

- What are Pseudo-classes?

A pseudo-class is used to define a special state of an element.

- For example, it can be used to:
 - I. Style an element when a user mouse is over it
 - II. Style visited and unvisited links differently
 - III. Style active links

```
selector:pseudo-class {  
  property: value;  
}
```

Pseudo-class Selectors Example

- Anchor Pseudo-classes
- Links can be displayed in different ways:
- **Note:** a:hover MUST come after a:link and a:visited in the CSS definition in order to be effective!
- **Note2:** a:active MUST come after a:hover in the CSS definition in order to be effective!

```
/* unvisited link */
a:link {
    color: #FF0000;
}

/* visited link */
a:visited {
    color: #00FF00;
}

/* mouse over link */
a:hover {
    color: #FF00FF;
}

/* selected link */
a:active #0000FF {
    color: ;
}
```

All Pseudo-class Selectors

Selector	Example	Example description
:active	a:active	Selects the active link
:checked	input:checked	Selects every checked <input> element
:disabled	input:disabled	Selects every disabled <input> element
:empty	p:empty	Selects every <p> element that has no children
:enabled	input:enabled	Selects every enabled <input> element
:first-child	p:first-child	Selects every <p> elements that is the first child of its parent
:first-of-type	p:first-of-type	Selects every <p> element that is the first <p> element of its parent
:focus	input:focus	Selects the <input> element that has focus
:hover	a:hover	Selects links on mouse over
:in-range	input:in-range	Selects <input> elements with a value within a specified range
:invalid	input:invalid	Selects all <input> elements with an invalid value
:lang(language)	p:lang(it)	Selects every <p> element with a lang attribute value starting with "it"
:last-child	p:last-child	Selects every <p> elements that is the last child of its parent
:last-of-type	p:last-of-type	Selects every <p> element that is the last <p> element of its parent
:link	a:link	Selects all unvisited links
:not(selector)	:not(p)	Selects every element that is not a <p> element

Pseudo-element Selectors

- What are Pseudo-Elements?

A CSS pseudo-element is used to style specified parts of an element.

- For example, it can be used to:
 - I. Style the first letter, or line, of an element
 - II. Insert content before, or after, the content of an element

```
selector::pseudo-element {  
  property: value;  
}
```

Example: The ::first-line Pseudo-element

- The ::first-line pseudo-element is used to add a special style to the first line of a text.
- The following example formats the first line of the text in all <p> elements:

```
p::first-line {  
  color: #ff0000;  
  font-variant: small-caps;  
}
```

- **Note:** The ::first-line pseudo-element can only be applied to block-level elements.

All Pseudo-Element Selectors

Selector	Example	Example description
::after	p::after	Insert content after every <p> element
::before	p::before	Insert content before every <p> element
::first-letter	p::first-letter	Selects the first letter of every <p> element
::first-line	p::first-line	Selects the first line of every <p> element
::selection	p::selection	Selects the portion of an element that is selected by a user

Attribute Selectors

- The [attribute] selector is used to select elements with a specified attribute.
- Example:

```
a[target] {  
  background-color: yellow;  
}
```

- selects all <a> elements with a target attribute.

Attribute Selectors

- The [attribute="value"] selector is used to select elements with a specified attribute and value.
- Example:

```
a[target="_blank"] {  
  background-color: yellow;  
}
```

- selects all <a> elements with a target="_blank" attribute

All CSS Attribute Selectors

Selector	Example	Example description
[attribute]	[target]	Selects all elements with a target attribute
[attribute=value]	[target=_blank]	Selects all elements with target="_blank"
[attribute~=value]	[title~=flower]	Selects all elements with a title attribute containing the word "flower"
[attribute =value]	[lang =en]	Selects all elements with a lang attribute value starting with "en"
[attribute^=value]	a[href^="https"]	Selects every <a> element whose href attribute value begins with "https"
[attribute\$=value]	a[href\$=".pdf"]	Selects every <a> element whose href attribute value ends with ".pdf"
[attribute*=value]	a[href*="BGU"]	Selects every <a> element whose href attribute value contains the substring "BGU"



Class Exercise 1- selectors



Comments in CSS

CSS Comments

- Comments are used to explain the code and may help when you edit the source code later.
- Comments are ignored by browsers.
- A CSS comment is placed inside the <style> element, and starts with /* and ends with */

```
/* This is a single-line comment */  
p {  
    color: red;  
}
```

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red; /* Set text color to red */
}
</style>
</head>
<body>

<h2>My Heading</h2>

<!-- These paragraphs will be red -->
<p>Hello World!</p>
<p>This paragraph is styled with CSS.</p>
<p>CSS comments are not shown in the
output.</p>

</body>
</html>
```



Colors

What Can Be Colored?

- Background `<h1 style="background-color:DodgerBlue;">Hello World</h1>`
- Text `<h1 style="color:Tomato;">Hello World</h1>`
- Border `<h1 style="border:2px solid Tomato;">Hello World</h1>`

```
h1 {  
  color: red;  
  background-color: blue;  
  border:2px solid Tomato;  
}
```

Colors

Colors can be referred by:

- Names-Red, Green, Blue...
- Values- RGB, RGBA, HSL, HSLA, HEX Code
- Variables

Color values

- RGB and RGBA

rgb(red, green, blue)

rgba(red, green, blue, alpha)

```
<h1 style="color:rgb(255, 99, 71);">...</h1>
```

```
<h1 style="color:rgba(255, 99, 71, 0.5);">...</h1>
```

- HEX Code

#rrggbb

```
<h1 style="color:#ff6347;">...</h1>
```

- HSL

hsl(hue, saturation, lightness)

```
<h1 style="color:hsl(9, 100%, 64%);">...</h1>
```

hsla(hue, saturation, lightness, alpha)

```
<h1 style="color:hsla(9, 100%, 64%, 0.5);">...</h1>
```

CSS Variables

- The var() function is used to insert the value of a CSS variable.
- CSS variables have access to the DOM.
- We can change the variables with JavaScript.
- Instead of copy and paste the same colors, you can place them in variables.

Syntax of the var() Function

- The var() function is used to insert the value of a CSS variable.
- The syntax of the var() function is as follows:

`--var(name, value)`

Value	Description
name	Required. The variable name (must start with two dashes)
value	Optional. The fallback value (used if the variable is not found)

var() Declaration

- CSS variables can be either **global** or **local**.
- Global variables:
 - can be accessed/used through the entire document.
- Local variable:
 - can be used only inside the selector where it is declared.
- To create a variable with global scope, declare it inside the :root selector. The :root selector matches the document's root element.
- To create a variable with local scope, declare it inside the selector that is going to use it.

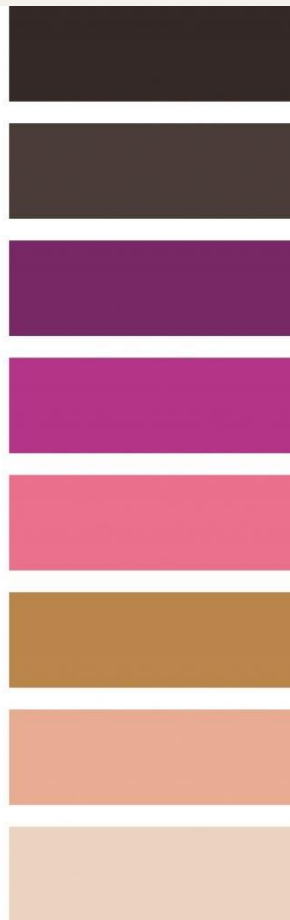
var() Usage

- First, we declare two global variables (--blue and --white).
- Then, we use the var() function to insert the value of the variables later in the style sheet:

```
:root {  
  --blue: #1e90ff;  
  --white: #ffffff;  
}  
  
body { background-color: var(--blue); }  
  
h2 { border-bottom: 2px solid var(--blue); }  
  
.container {  
  color: var(--blue);  
  background-color: var(--white);  
  padding: 15px;  
}  
  
button {  
  background-color: var(--white);  
  color: var(--blue);  
  border: 1px solid var(--blue);  
  padding: 5px;  
}
```

Color Scheme

- A good color scheme is critical when designing a web page.
- Color is a massive aspect of branding → select your color scheme in accordance .
- Choose a primary color, additional color, background color and typeface color (color of text).
- Use online resources and tools for help and inspiration:
 - <https://coolors.co/>
 - <https://developer.mozilla.org/en-US/docs/Web/CSS/named-color>



Color Gradient

- CSS gradients let you display smooth transitions between two or more specified colors.
- CSS defines two types of gradients:
 - I. Linear Gradients (goes down/up/left/right/diagonally)
 - II. Radial Gradients (defined by their center)

```
body{  
background-image: linear-gradient(to left, #93c572, #f5f2d0);  
}
```



CSS



Text

Text Properties

- Color- the color of the text
- Alignment- right, left, center, justify etc.
- Decoration- underline, line through, overline etc.
- Transformation- transform text to uppercase / lowercase etc.
- Spacing- add space between letters
- shadow- add shadow to the text

Font Properties

- Font-style- normal / *italic*
- Font-variant- normal / small-caps (M_Y N_AM_E I_S I_TZI_K)
- Font-weight- bold / bolder / lighter etc.
- Font-size/line-height- the size of the font.
- Font-family- the font you use (Ariel, Times New Roman etc.)

Font Family

- The font family is the font we'll use.
- We will always choose one more font-family, so that if the OS doesn't support the chosen font-family, it will be displayed with the defaulted font-family we specified.
 - `font-family: Arial, Helvetica, "sans-serif";`
 - in this example, the font is Ariel. If it can't be read by the OS- Helvetica will be chosen.
- For fonts with more than one word in its name, we will use ""

CSS Web Fonts using fonts.google.com

- If you do not want to use any of the standard fonts in HTML, you can use Google Fonts.
- Google Fonts are free to use and have more than 1000 fonts to choose from.
 - Go to fonts.google.com
 - Choose the font you like
 - Style the font according to your preferences
 - Copy the link to the font and the CSS rule
 - add the link to the <head> element
 - Add the CSS rule to the elements on which you wish to apply the font

Web Font Example



Google Fonts

Search fonts

Sort by: Trending



Specimen

Type tester

Glyphs

About & license

Download family

Styles

Type here to preview text

48px



Regular 400

Whereas recognition of the inherent dignity

Remove Regular 400

Regular 400

Regular 400 at 48px

Whereas a common understanding
of these rights and freedoms is

Regular 400 at 36px

No one shall be held in slavery or servitude;

Regular 400 at 21px

No one shall be subjected to arbitrary arrest, detention or exile. Everyone is entitled in full equality to a fair and public hearing by an independent and impartial tribunal, in the determination of his rights and obligations and of any criminal charge against him. No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.

Selected family



Review

Pacifico



Regular 400



Add more styles Remove all

Use on the web

To embed a font, copy the code into the <head> of your html

☒ <link> ☐ @import

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Pacifico&display=swap" rel="stylesheet">
```



CSS rules to specify families

```
font-family: 'Pacifico', cursive;
```



[Read our FAQs](#)



Sizes

CSS Height and Width

- The height and width properties are used to set the height and width of an element.
- The height and width properties may have the following values:
 - I. auto - This is default. The browser calculates the height and width
 - II. length - Defines the height/width in px, cm etc.
 - III. % - Defines the height/width in percent of the containing block
 - IV. initial - Sets the height/width to its default value
 - V. inherit - The height/width will be inherited from its parent value

Absolute Lengths

Unit	Description
cm	Centimeters
mm	Millimeters
in	inches (1in = 96px = 2.54cm)
px *	pixels (1px = 1/96th of 1in = 0.26 mm)
pt	points (1pt = 1/72 of 1in = 0.35 mm)
pc	picas (1pc = 12 pt)

Relative Lengths

Unit	Description
em	Relative to the font-size of the father element (2em means 2 x font size of father)
rem	Relative to font-size of the root element
ch	Relative to the width of the "0" (zero)
vw	Relative to 1% of the width of the viewport*
vh	Relative to 1% of the height of the viewport*
vmin	Relative to 1% of viewport's* smaller dimension
vmax	Relative to 1% of viewport's* larger dimension
%	Relative to the parent element

```
<!DOCTYPE html>
<html lang="en" style="font-size: 100%;">
<head>
```

```
  <meta charset="UTF-8">
  <title>sizes</title>
  <style>
    body {
      font-size: 100%;
    }

    h1 {
      font-size: 2.5em;
    }

    h2 {
      font-size: 2em;
    }

    p {
      font-size: 0.875rem;
    }
  </style>
```

```
</head>
<body>
  <h1>This is heading 1</h1>
  <h2>This is heading 2</h2>
  <p>This is a paragraph.</p>
  <p>Specifying the font-size in percent and em displays
the same size in all major browsers, and allows all
browsers to resize the text!</p>
</body>
</html>
```

This is heading 1

This is heading 2

This is a paragraph.

Specifying the font-size in percent and em displays the same size in all major browsers, and allows all browsers to resize the text!